David Xia

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Education

- **University of Illinois Urbana-Champaign** *August 2023 - May 2027*
  Bachelor of Science in Computer Science Champaign, Illinois
  Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Math Doctoral Preparation Concentration
  - GPA: 3.98/4.00

Experience

- **Illinois Combinatorics Lab for Undergraduate Experience (ICLUE) [🌐]**, Researcher *Jan 2024 - Present*
  - Mentored by Dr. Alexander Yong and doctoral student Duy Phan, studying algebraic combinatorics
  - Currently researching the application of the Robinson-Schensted-Knuth correspondence for bounding the probability of specific interactions in the Schensted insertion of permutations
- **National University of Singapore SEEDER Group [🌐]**, Researcher *May 2024 - July 2024*
  - Worked under guidance of Dr. Kelvin Xuanyao's SEEDER Group to developed GARDEN, a machine learning pipeline to detect, classify, and analyze plant diseases, improving previous accuracies by 40%
  - Implemented generative, classification, and segmentation models built utilizing PyTorch and TensorFlow framework, using self-collected data from local aquaculture farms, and presented findings in manuscript
- **University of California San Diego**, Research Intern *June 2020 - August 2023*
  - Developed EveGuard, a software framework that defends against smartphone side-channel eavesdropping using an adversarial perturbation generator model while maintaining audio quality
  - Trained segmentation and classification models for CogTeach, a facial expression recognition application tool for virtual learning, and conducted pilot study by delivering courses of lectures on neural networks

Projects

- **Predictability and Competitiveness of Sports Leagues**, Python [GitHub] [GitHub] *January 2024 - Present*
  - Researched the data science perspective of sports with Dr. AJ Hildebrand at Illinois Mathematics Lab
  - Built web scrapers with Selenium to scrape online data and analyzed the predictability of sports matches by comparing prediction accuracies of multiple sources including polls and the betting market
  - Evaluated efficiency of model based probabilistic predictions
- **GARDEN: ML pipeline for anomaly detection**, PyTorch, TensorFlow [GitHub] *May 2024 - July 2024*
  - Developed machine learning pipeline to detect, classify, and analyze diseases found in plant leaves
  - Trained U-Net segmentation models and classification models based on convolutional neural networks to isolate and classify images using the TensorFlow framework, achieving 90% segmented classification rate
  - Generated synthetic images using custom trained cGAN model to increase data variety and volume
  - Stacked deep learning models with traditional techniques like support vector machines, gradient boosting, logistic regression, and decision trees to achieve an increased detection accuracy of over 95%
- **EveGuard: Software framework for eavesdropping defense**, TensorFlow, GAN [🌐] *May 2022 - August 2023*
  - Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality
  - Implemented Eve-GAN, a translation task for inferring eavesdropped signals, enabling training of PGM
  - Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.

Presentations

2024 **Rose-Hulman Undergraduate Math Conference** Predictability in College Sports

2024 **University of Illinois Undergraduate Research Symposium** Predictability in College Sports

2024 **Illinois Mathematics Lab Research Symposium** Predictability in College Sports

2025 **Joint Mathematics Meeting Pi Mu Epsilon Poster Session** Predicatbility in Major Sports Leagues

Papers

PRE=Preprint, ACC=Accepted

[1] **[ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations** [arXiv]
Jung-Woo Chang, Ke Sun, **David Xia**, Xinyu Zhang, Farinaz Koushanfar
To appear in IEEE Symposium on Security and Privacy, 2025

Skills

• Python • C++ • PyTorch • TensorFlow • MatLab • NumPy • SciPy • Matplotlib • Pandas • Sage • HuggingFace •